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DEPARTMENT OF ELECTRICAL ENGINEERING AND COMPUTER SCIENCE

EE 516 Power Systems

Final Project: Power Flow Analysis Using Gauss–Seidel Method and MIT Formulation

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1. Introduction

This project focuses on solving power system load flow problems using numerical techniques, primarily the Gauss–Seidel method combined with the MIT bus admittance formulation. The goal is to determine bus voltages, power injections, and line flows under different operating conditions.

Many of the computations are performed in per-unit (pu) and some in (MW/MVAR), which simplifies calculations and ensures consistency across different voltage levels.

2. Objectives

The main objectives of this project are:

- To implement the Gauss–Seidel method for solving load flow problems
- To construct the bus admittance matrix using line data and MIT formulation
- To compute bus voltages, power injections, and line flows
- To analyze the effect of reactive power injection on system performance
- To study the impact of network changes such as line removal
- To verify results using standard benchmark examples

3. Problem-Solving Approach

The solution approach follows a structured process:

First, the system is modeled using per-unit quantities. The bus admittance matrix Y_{bus} is constructed either directly from line parameters or using the MIT formulation:

$$Y_{bus}=AY_LA^T$$

Next, buses are classified as slack, PV, or PQ buses. Initial voltage values are assumed, and the Gauss–Seidel method is applied iteratively.

For each iteration:

- PQ bus voltages are updated using power equations
- PV bus reactive power is calculated, while maintaining fixed voltage magnitude
- The slack bus remains unchanged

The process continues until the voltage difference between iterations is below a specified tolerance, ensuring convergence.

Finally, bus power injections, line flows, and losses are computed from the converged voltages.

4. Simulation

MATLAB was used to implement the Gauss–Seidel load flow algorithm. The simulation includes:

- Construction of the Y_{bus} matrix
- Iterative voltage updates
- Handling of PV and slack buses
- Calculation of power injections and line currents

Three cases were analyzed for the six-bus system:

- Base case
- Reactive power injection at Bus 5
- Line removal scenario

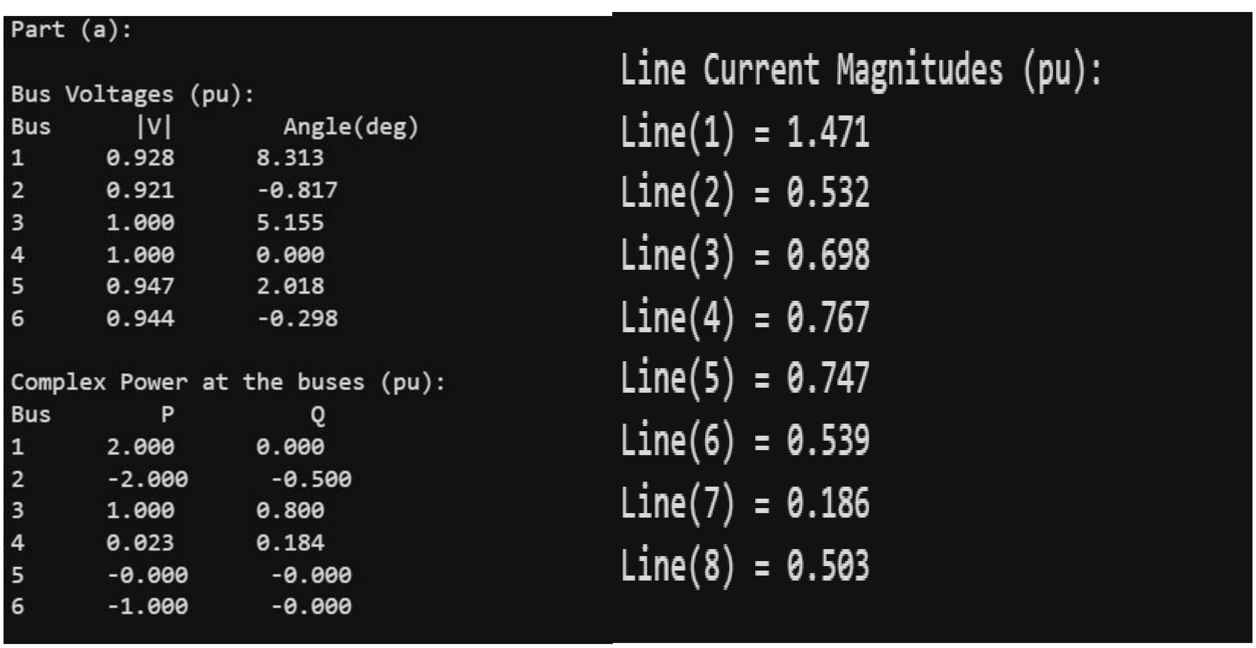
Problems 2 and 3 were simulated using 3-bus systems to validate results against known solutions.

5. Results

Problem 1

The system converged successfully for all cases.

- In the base case, voltages remain within acceptable limits, showing normal operation.
- Injecting reactive power at Bus 5 improves voltage levels and reduces system stress.
- Removing a transmission line causes voltage drops and increases current in remaining lines, indicating higher loading conditions.



```
Part (a):

Bus Voltages (pu):
Bus      |V|       Angle(deg)
1      0.928      8.313
2      0.921      -0.817
3      1.000      5.155
4      1.000      0.000
5      0.947      2.018
6      0.944      -0.298

Complex Power at the buses (pu):
Bus        P          Q
1      2.000      0.000
2      -2.000      -0.500
3      1.000      0.800
4      0.023      0.184
5      -0.000      -0.000
6      -1.000      -0.000
```

```
Line Current Magnitudes (pu):
Line(1) = 1.471
Line(2) = 0.532
Line(3) = 0.698
Line(4) = 0.767
Line(5) = 0.747
Line(6) = 0.539
Line(7) = 0.186
Line(8) = 0.503
```

Figure 1. Part (a): Base case matlab outputs

```
Part (b): Q = 0.5 pu injected at Bus 5

Bus Voltages (pu):
Bus      |V|        Angle(deg)
1      0.961      7.740
2      0.952      -0.782
3      1.000      5.016
4      1.000      0.000
5      0.981      1.872
6      0.971      -0.291

Complex Power at the buses (pu):
Bus        P          Q
1      2.000      0.000
2      -2.000      -0.500
3      1.000      0.325
4      0.020      0.094
5      -0.000      0.500
6      -1.000      -0.000
```

```
Line Current Magnitudes (pu):
Line(1) = 1.422
Line(2) = 0.534
Line(3) = 0.676
Line(4) = 0.764
Line(5) = 0.573
Line(6) = 0.478
Line(7) = 0.096
Line(8) = 0.425
```

Figure 2. Part (c): Q = 0.5 pu injected at Bus matlab outputs

```
Part (c): Line 1 Removed

Bus Voltages (pu):
Bus      |V|        Angle(deg)
1      0.818      28.186
2      0.859      -4.363
3      1.000      6.107
4      1.000      0.000
5      0.885      3.694
6      0.893      -1.058

Complex Power at the buses (pu):
Bus        P          Q
1      2.000      0.000
2      -2.000      -0.500
3      1.000      1.693
4      0.067      0.356
5      -0.000      -0.000
6      -1.000      -0.000
```

```
Line Current Magnitudes (pu):
Line(1) = 1.246
Line(2) = 2.446
Line(3) = 1.481
Line(4) = 1.214
Line(5) = 0.798
Line(6) = 0.362
Line(7) = 1.207
```

Figure 3. Part (c): Line 1 Removed matlab outputs

Problem 2 (Example 7)

The results confirm that the slack bus supplies the required real and reactive power to meet system demand. Power flows from the source toward the loads, and line losses are observed as expected.
The computed values match the reference solution, validating the implementation.

```
Problem 2 - Example 7 Results
Converged in 21 iterations

Bus Voltages(pu):
Bus      |V|        Angle(deg)
1      1.05000      0.0000
2      0.98184      -3.5035
3      1.00125      -2.8624

Bus Power Injections(pu):
Bus        P          Q
1      4.095      1.890
2      -2.566      -1.102
3      -1.386      -0.452
```

```
Line Flows and Losses (MW/MVAR)

S12 = 199.50 + j84.00
S21 = -191.00 - j67.00
SL12 = 8.50 + j17.00
S13 = 210.00 + j105.00
S31 = -205.00 - j90.00
SL13 = 5.00 + j15.00
S23 = -65.60 - j43.20
S32 = 66.40 + j44.80
SL23 = 0.80 + j1.60
```

Figure 4) Problem 2/Example 7 using MIT formulation matlab outputs

Problem 3 (Example 8)

The presence of a PV bus demonstrates voltage control through reactive power adjustment.

- The PV bus maintains its voltage magnitude by varying reactive power output
- The slack bus balances the remaining power
- Line flows and losses are consistent with system behavior

```
Problem 3 - Example 8 Results
Converged in 22 iterations

Bus Voltages:
Bus      |V|        Angle(deg)
1      1.05000      0.0000
2      0.97168      -2.6965
3      1.04000      -0.4988

Bus Power Injections:
Bus        P          Q
1      2.18423    1.40852
2      -4.00000    -2.50000
3      2.00000    1.46177
```

```
Line Flows and Losses (MW/MVAR)

S12 = 179.36 + j118.734
S21 = -170.97 - j101.947
SL12 = 8.39 + j16.787
S13 = 39.06 + j22.118
S31 = -38.88 - j21.569
SL13 = 0.18 + j0.548
S23 = -229.03 - j148.053
S32 = 238.88 + j167.746
SL23 = 9.85 + j19.693
```

Figure 5) Problem 3-Example 8 using MIT formulation matlab outputs

6. Conclusions

This project demonstrated the application of both the Gauss–Seidel method and the MIT formulation in solving power flow problems. The MIT formulation provided a systematic way to construct the bus admittance matrix, while the Gauss–Seidel method was used to iteratively compute the bus voltages.

The results show that the Gauss–Seidel method is effective for obtaining accurate solutions, although it may require a relatively high number of iterations depending on the tolerance. The MIT formulation simplifies network modeling and ensures a consistent representation of system connectivity.

The analysis also highlights the importance of reactive power in maintaining voltage levels and system stability. Changes such as reactive power injection and line removal significantly affect voltage profiles and power distribution across the network.

7. Appendix

The MATLAB codes for all three problems.

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EE516Finalprojectqn1.m
EE516Finalprojectqn1.m
EE516Finalprojectqn1.m
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